



**INTERNATIONAL CAMPUS
OF SCIENCE AND TECHNOLOGY**

International Campus of Science and Technology

Faculty of Computing, Department of IT
HDIT Batch 01

HDIT - DBMS

Project Name: Restaurant Management System - Cafulon

Submitted by:

1. NM. Roshaan Akther
2. A. Washid
3. MAM. Masheef
4. MIF. Fahmidha
5. MNF. Nuha
6. WMCS. Weerakkodi

Submission Date: Jun 9, 2026

NORMALIZATION Restaurant Order Processing

UNF (Unnormalized Form)

A single order record containing all data:

```
Orders_UNF(  
  OrderID,  
  OrderDate,  
  CustomerName,  
  CustomerPhone,  
  CustomerEmail,  
  EmployeeName,  
  EmployeePosition,  
  TableNumber,  
  (ItemName, CategoryName, Quantity, UnitPrice, Subtotal),  
  TotalAmount,  
  PaymentMethod,  
  PaymentAmount  
)
```

Sample UNF Record:

```
(101, 2025-06-01, 'Nimal Perera', '071-2345678', 'nimal@email.com',  
'Ali Raza', 'Manager', 3,  
(('Bruschetta', 'Appetizers', 2, 8.50, 17.00,  
'Grilled Chicken', 'Main Course - Chicken', 1, 16.00, 16.00,  
'Fresh Orange Juice', 'Beverages', 1, 5.00, 5.00),  
38.50, 'Credit Card', 38.50)
```

Problems:

- Repeating group (multiple items per order)
- Data redundancy (customer/employee info repeated for every order)
- Insert/update/delete anomalies

1NF (First Normal Form)

Remove repeating groups — create separate rows for each item. Add a composite key.

```
Order_1NF(  
  OrderID,      -- PK (partial)  
  ItemName,     -- PK (partial)  
  OrderDate,  
  CustomerName,  
  CustomerPhone,  
  CustomerEmail,  
  EmployeeName,  
  EmployeePosition,  
  TableNumber,  
  CategoryName,  
  Quantity,  
  UnitPrice,  
  Subtotal,  
  TotalAmount,  
  PaymentMethod,  
  PaymentAmount  
)
```

Sample 1NF Data:

OrderID	ItemName	OrderDate	CustomerName	..	Quantity	UnitPrice	Subtotal
101	Bruschetta	2025-06-01	Ahmed Khan	..	2	8.50	17.00
101	Grilled Chicken	2025-06-01	Ahmed Khan	..	1	16.00	16.00
101	Fresh Orange Juice	2025-06-01	Ahmed Khan	..	1	5.00	5.00

Key: Composite (OrderID, ItemName)

Problems remaining: Partial dependencies — CustomerName depends only on OrderID, not on ItemName. Same for EmployeeName, TotalAmount, etc.

2NF (Second Normal Form)

Remove partial dependencies — create separate tables for entities that depend only on part of the composite key.

Table 1: Order (OrderID → all order-level fields)

```
Order_2NF(  
  OrderID,      -- PK  
  OrderDate,  
  CustomerName,  
  CustomerPhone,  
  CustomerEmail,  
  EmployeeName,  
  EmployeePosition,  
  TableNumber,  
  TotalAmount,  
  PaymentMethod,  
  PaymentAmount  
)
```

Table 2: OrderItem (OrderID + ItemName → item-level fields)

```
OrderItem_2NF(  
  OrderID,      -- PK (FK → Order)  
  ItemName,     -- PK  
  CategoryName,  
  Quantity,  
  UnitPrice,  
  Subtotal  
)
```

Problems remaining: Transitive dependencies — CustomerName → CustomerPhone, CustomerEmail. EmployeeName → EmployeePosition. ItemName → CategoryName.

3NF (Third Normal Form)

Remove transitive dependencies — separate customer, employee, menu, and payment into their own tables.

Final Normalized Schema:

```
Customer(  
  CustomerID,      -- PK  
  FirstName,  
  LastName,  
  Phone,  
  Email,  
  Address,  
  City,  
  RegistrationDate,  
  IsVIP  
)
```

```
Employee(  
  EmployeeID,      -- PK  
  FirstName,  
  LastName,  
  Position,  
  Phone,  
  Email,  
  Salary,  
  HireDate,  
  Shift  
)
```

```
MenuCategory(  
  CategoryID,      -- PK  
  CategoryName,  
  Description  
)
```

```
MenuItem(  
  ItemID,          -- PK  
  CategoryID,      -- FK → MenuCategory  
  ItemName,  
  Description,  
  Price,  
  IsAvailable,  
  PreparationTime  
)
```

```
RestaurantTable(  
    TableID,          -- PK  
    TableNumber,  
    Capacity,  
    Location,  
    IsActive  
)
```

```
Reservation(  
    ReservationID,     -- PK  
    CustomerID,        -- FK → Customer  
    TableID,           -- FK → RestaurantTable  
    EmployeeID,        -- FK → Employee  
    ReservationDate,  
    ReservationTime,  
    PartySize,  
    Status,  
    SpecialRequests,  
    CreatedAt  
)
```

```
Order(  
    OrderID,           -- PK  
    CustomerID,        -- FK → Customer  
    EmployeeID,        -- FK → Employee  
    TableID,           -- FK → RestaurantTable  
    OrderDate,  
    OrderType,  
    Status,  
    TotalAmount  
)
```

```
OrderDetail(  
    OrderDetailID,     -- PK  
    OrderID,           -- FK → Order  
    ItemID,            -- FK → MenuItem  
    Quantity,  
    UnitPrice,  
    Subtotal,  
    Notes  
)
```

```
Payment(  
    PaymentID,         -- PK  
    OrderID,           -- FK → Order  
    PaymentDate,  
    Amount,  
    PaymentMethod,  
    TransactionReference,
```

IsPaid
)

Normalization Summary

Normal Form	Status	Key Change
UNF	Repeating groups	Single table with multi-valued attributes
1NF	Atomic values	Removed repeating groups, composite key
2NF	No partial dependencies	Separated order-level from item-level data
3NF	No transitive dependencies	Separated customer, employee, menu, payment entities

All tables in the final schema satisfy 3NF — every non-key attribute is:

1. Atomic (1NF) ✓
2. Fully functionally dependent on the primary key (2NF) ✓
3. Non-transitively dependent on the primary key (3NF) ✓